

DS Semester Examination 2024

Data Structures (DS) · Semester Examination · 2024

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Compare Asymptotic Notation with Sparse Matrices. When would you prefer one over the other? (5 marks)
2. Explain Stack ADT with a neat diagram and a suitable example. (5 marks)
3. Discuss the role of Binary Tree Traversals in Trees & Binary Search Trees. (5 marks)
4. Discuss the role of Graph Representations in Graphs & Graph Algorithms. (5 marks)
5. Discuss the role of Binary Search in Sorting, Searching & Hashing. (5 marks)

Part B

1. Compare Asymptotic Notation with Arrays. When would you prefer one over the other? (Unit 1)
2. What is meant by Arrays? Discuss its advantages and limitations. (Unit 1)
3. Write short notes on Stack ADT. (Unit 2)
4. Define Infix to Postfix Conversion. Derive or justify the key properties involved. (Unit 2)
5. Discuss the role of Binary Tree Traversals in Trees & Binary Search Trees. (Unit 3)
6. Discuss the role of Binary Search Trees in Trees & Binary Search Trees. (Unit 3)
7. Compare Graph Representations with Topological Sorting. When would you prefer one over the other? (Unit 4)
8. Explain Breadth-First Search with a neat diagram and a suitable example. (Unit 4)
9. What is meant by Binary Search? Discuss its advantages and limitations. (Unit 5)
10. Discuss the role of Quick Sort in Sorting, Searching & Hashing. (Unit 5)